

Beyond the Given Wave Equation

From the RCP/LCP $A_{z2,3}$ ODE (khaxn.pdf) to a validated theory
of the non-Abelian Kelvin–Helmholtz instability, and its confrontation with GPU
data

SU(2) Yang–Mills KH simulation program

2026-07-21

Where we left off

Already presented

khaxn.pdf: linearized Yang–Mills–Wong system in the $a_x^a = 0$ gauge. Its eq. (23) closes the perturbation into a single decoupled 2nd-order wave equation for the circular combination a_z^+ (i.e. $A_{z2} \pm iA_{z3}$):

$$\left(-\partial_x^2 - \omega^2 + k_z^2 + \alpha^2(\phi_0^1)^2 - \alpha^2(A_{0z}^1)^2\right)a_z^+ = \frac{\alpha^2 V_{0z}}{\Omega + \alpha\phi_0^1} \left(V_{0z} - \frac{k_z + \alpha A_{0z}^1}{\omega + \alpha\phi_0^1}\right)a_z^+$$

This talk covers everything derived and measured since. Two more given results are used silently downstream — restated briefly next — then it's all new: the quartic → exact scalar reduction → growth ceiling → exact-action quantization → confinement-set wavelength selection → full GPU validation campaign → where Letter A actually stands today.

The second given result (briefly): wkb.pdf

Expanding the step-potential form of the boxed equation to $\mathcal{O}(v^2)$ gives a Schrödinger form $\partial_x^2 a_z + Q_{\pm}(x)a_z = 0$ with a *parabolic* well,

$$Q_{\pm}(x) = Q_{0,\text{mean}} \pm \delta - |C_1|x^2 + C_2x^4, \quad |C_1| = \frac{\alpha^3 v^2}{2\omega^2},$$

and Bohr–Sommerfeld quantization of that harmonic core (the $\pm\delta$ jump cancels between the two half-integrals) gives wkb.pdf eq. (33):

$$\boxed{Q_{0,\text{mean}} = (2n + 1)\sqrt{|C_1|}} \quad n = 0, 1, 2, \dots$$

Physical picture

This is a harmonic-oscillator quantization: exact at $k_z = 0$, an uncontrolled extrapolation away from it. Everything from here derives *when* that extrapolation works, replaces it where it doesn't, and confronts both against real GPU data.

Outline

- 1 From eq. (33) to the quartic, $k_z=0$ anchor
- 2 The time-domain model
- 3 Six-field linearization
- 4 Local dispersion: tachyonic & drive branch
- 5 Exact scalar reduction: one ODE for $a(x)$
- 6 The drive, resonance, growth ceiling
- 7 Quantization: models A and B
- 8 The dispersion curve: confinement-set peak
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- 17 Where this leaves Letter A

Clearing eq. (33) to a quartic

Substituting the harmonic-core definitions into eq. (33) and writing $C \equiv (2n+1)\sqrt{\alpha^3/2} V_0$:

$$\omega^2 - k_z^2 - \frac{\alpha^2 V_0 k_z}{\omega^2} = \frac{C}{\omega}$$

Multiply through by ω^2 ($\omega = 0$ is not a root) and collect:

$$\omega^4 - k_z^2 \omega^2 - C\omega - \alpha^2 V_0 k_z = 0$$

the **WKB quartic**. $\gamma = \max \text{Im } \omega$ over its four roots is the “ γ_{WKB} ” quoted in every early program table.

Caveat

An early implementation assumed $\text{Re } \omega = 0$ (purely growing) and bisected on real γ — wrong away from $k_z = 0$ by up to 50% at $k_z = 0.25$. The full complex quartic must be solved.

The $k_z = 0$ limit: chromo-Weibel growth rate

Set $k_z = 0$: $\omega^4 - C\omega = 0 \Rightarrow \omega(\omega^3 - C) = 0$. The growing root is $\omega = C^{1/3} e^{2\pi i/3}$, giving

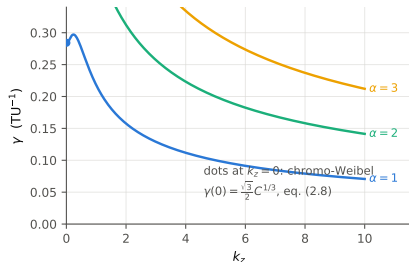
$$\gamma_{\text{Weibel}}(k_z=0) = \left(\sqrt{\frac{\alpha^3}{2}} V_0 \right)^{1/3} \sin 60^\circ$$

Physical picture

Not a bunching instability — nothing varies along z . It's *filamentation*: A_{z1} rotates color-2/3 fields into each other while counter-streaming currents feed them back. Closing the loop $E \rightarrow A \rightarrow Q \rightarrow J \rightarrow E$ gives gain $\propto C/\gamma^3$; marginal self-amplification fixes $\gamma \sim C^{1/3}$, root geometry supplies $\sin 60^\circ$.

Validated to 0.5%: $\alpha = 2$, $V_0 = 0.1$: $\gamma_{\text{sim}} = 0.5039$ vs. formula 0.5065, over nine decades of growth — the strongest single anchor of the quartic, and the first number quoted to a referee (parameter-free).

Quartic vs. reality: where it's valid, where it fails



- $k_z = 0$: exact, validated to 0.5%.
- **Low** k_z : the quartic's drive $\alpha^2 V_0 k_z / \gamma^2$ has no ceiling $\Rightarrow$ overshoots the true branch by 10–20 $\times$.
- **High** k_z : quartic balance gives $\gamma^2 \approx \alpha^2 V_0 / k_z$, a decline, while the true branch saturates near a ceiling $\Rightarrow$ $\sim 25\%$ underestimate.

The diagnosis (why, precisely) has to wait for the exact reduction — coming up.

The simulated system

For each color $a \in \{1, 2, 3\}$: fields $E_x^a, E_z^a, B_y^a, A_z^a$. For each beam $s \in \{A, B\}$: fluid n_s, p_{xs}, p_{zs} , color charge Q_s^a .

$$\begin{aligned}\partial_t n + \nabla \cdot (n \vec{v}) &= 0, & \partial_t p_x + \nabla \cdot (\vec{v} p_x) &= -n \sum_a Q^a (E_x^a + v_z B_y^a) \\ \partial_t p_z + \nabla \cdot (\vec{v} p_z) &= -n \sum_a Q^a (E_z^a - v_x B_y^a), & \partial_t Q^a &= \alpha v_z (\vec{Q} \times \vec{A}_z)^a \\ \partial_t E_x^a &= -\partial_z B_y^a - \alpha (\vec{A}_z \times \vec{B}_y)^a - J_x^a, & \partial_t E_z^a &= \partial_x B_y^a - J_z^a \\ \partial_t B_y^a &= \partial_x E_z^a - \partial_z E_x^a + \alpha (\vec{A}_z \times \vec{E}_x)^a, & \partial_t A_z^a &= -E_z^a\end{aligned}$$

Beams: $v_{zA} = +V_0 \tanh \xi$, $Q_A^1 = +1$; $v_{zB} = -V_0 \tanh \xi$, $Q_B^1 = -1$; frozen background $A_{z1}(\xi) = -V_0 \ln \cosh \xi$, $u(\xi) \equiv -\alpha A_{z1}(\xi) \geq 0$.

A gap worth closing: gauge counting

SU(2) gauge freedom eliminates exactly *one* full spacetime component of A_μ^a — not two. Dropping both A_x^a, A_y^a while also fixing A_0^a needs justification, not assumption.

one gauge choice $\Rightarrow$ one component eliminated — not two.

- `khaxn.pdf` spends its choice on $a_x^a = 0$; ϕ survives, killed only by the Gauss *constraint*; a_y survives, decoupled from a_z by construction.
- This model spends a *different* choice, $A_0^a \equiv 0$ (temporal gauge — exactly what $\partial_t A_z = -E_z$ states).
- A_y is then a *sector* choice (never populated); A_x is wired to *nothing* in any equation of motion — a complete spectator either way.

Physical picture

One knob, one turn: two different single choices, not one spent twice. E_x carries exactly ϕ 's slaved role ($\gamma \mathbf{e}_x = -i\Omega_A \mathbf{b}$, purely algebraic).

Precession source from Wong's equation

Wong's equation, non-relativistic limit, temporal gauge, only $A_z^b \neq 0$:

$$\frac{dQ^a}{dt} = -g\epsilon^{abc}v_z A_z^b Q^c = +\alpha v_z (\vec{Q} \times \vec{A}_z)^a$$

exactly the model's source; $|\vec{Q}|$ conserved $\Rightarrow Q_{A,B}^1 = \pm 1$ stays a unit adjoint vector. **Energy budget** (derivation skipped): non-Abelian cross terms do *not* cancel —

$$\left. \frac{dE}{dt} \right|_{\text{NA}} = -2\alpha \int \vec{A}_z \cdot (\vec{B}_y \times \vec{E}_x)$$

Physical picture

A frozen field is an infinite battery. In the tachyonic region this term *feeds* the wave without limit (Nielsen–Olesen-type decay). In the shear well it's a drain; the shear branch draws on the beams' kinetic energy instead — a finite, self-consistent reservoir.

Circular pairs: color rotation is a phase

Combine color-2/3 perturbations: $b = B_y^2 + iB_y^3$, $e_x = E_x^2 + iE_x^3$, $e_z = E_z^2 + iE_z^3$, $a = A_z^2 + iA_z^3$, $q_{A,B} = Q_{A,B}^2 + iQ_{A,B}^3$, ansatz $\propto e^{ik_z z + \gamma t}$. For background $\vec{A}_z = (A_{z1}, 0, 0)$, any commutator $(\vec{A}_z \times \vec{X})^2 + i(\vec{A}_z \times \vec{X})^3 = iA_{z1}X^c$: a cross product with the background acts on a circular pair as multiplication by iA_{z1} .

$$\Omega_A(\xi) = k_z + \alpha A_{z1} = k_z - u(\xi), \quad \Omega_F(\xi) = k_z - \alpha A_{z1} = k_z + u(\xi)$$

Physical picture

The color-2/3 waves are *charged* (charge ± 1) under the residual $U(1)$ about the color-1 axis; αA_{z1} shifts their momentum like an ordinary gauge potential, $k_z \rightarrow k_z \pm \alpha A_{z1}$. Ampère sees Ω_A , Faraday sees Ω_F — the split is the fingerprint of the fixed $A_x = 0$ constraint.

The six-field eigenvalue system

$$\begin{array}{ll} \gamma b = -i\Omega_F e_x + e'_z & \gamma a = -e_z \\ \gamma e_x = -i\Omega_A b & \gamma q_A = i\alpha v a - i v \Omega_A q_A \\ \gamma e_z = b' + v(q_A - q_B) & \gamma q_B = i\alpha v a + i v \Omega_A q_B \end{array}$$

This is exactly what `ym_eigenmode.py` solves. Validated against the 2D simulation at 1–4% across the production grid (6–10% archive-wide clean core). **Correspondence check:** eliminating e_x, e_z, q_A, q_B, b reproduces exactly the beam-response denominators $\Omega \pm \alpha \Phi_0^1$ of the boxed equation on slide 2, under $\omega = i\gamma$ — the time-domain model linearizes into *precisely* the given frequency-domain theory, modulo two stated scope truncations (no background ϕ_0^1 ; no $a_y^\pm$ sector). **Symmetry:** helicity conjugation ($A_{z1} \rightarrow -A_{z1}$) combined with $\xi \rightarrow -\xi$ maps the system onto itself — every eigenmode on one side of the layer has a mirror partner of the opposite polarization on the other side, same γ .

Block factorization at $k_x = 0$

Freezing the background locally, the system splits into two independent blocks:

$$\underbrace{\gamma b = -i\Omega_F e_x, \quad \gamma e_x = -i\Omega_A b}_{\text{EM block}}$$

$$\underbrace{\gamma e_z = v(q_A - q_B), \quad \gamma a = -e_z, \quad \gamma q_{A,B} = i\alpha v a \mp i v \Omega_A q_{A,B}}_{\text{charge block}}$$

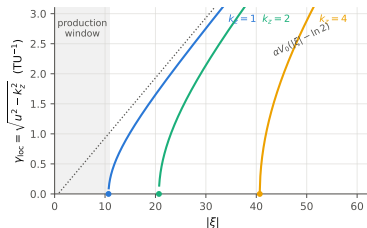
EM block: the tachyonic (Nielsen–Olesen) branch

Multiplying the EM block:

$$\gamma^2 = -\Omega_A \Omega_F = \alpha^2 A^2 - k_z^2$$

Negative wherever the background shift exceeds the wavenumber $\Rightarrow \gamma^2 > 0$: the charged-vector-on-strong-background instability of the Nielsen–Olesen class.

$$\gamma_{\text{loc}}(\xi) \simeq \sqrt{(\alpha V_0)^2 (|\xi| - \ln 2)^2 - k_z^2} \quad \text{real for} \quad |\xi| > \xi_{\text{crit}} \simeq \frac{k_z}{\alpha V_0} + \ln 2$$



Consequence: high- α window collapse

A window of radius ξ_w excludes the tachyon only if $\xi_w < \xi_{\text{crit}}$. As α grows at fixed k_z, V_0 : $\xi_{\text{crit}} \rightarrow \ln 2$ — eventually *no* window both contains the shear well and excludes the tachyon.

Caveat

This is the mechanism of the early tachyonic blow-ups at $\alpha \geq 6$: at $\alpha = 6, V_0 = 0.05, k_z = 2$: $\xi_{\text{crit}} = 7.4$, already inside the vetted windows. The failure maps to α , exactly as predicted — this is now known to be only *part* of the story (resolution addendum, later).

Same mechanism, evaluated at layer scale ($|A_{z1}| \sim V_0$) rather than the outer plateau, gives the “precession cascade” $\gamma_{\text{casc}} \approx \alpha V_0$ that motivated full eigenmode seeding (later in this talk).

Charge block: the local drive branch

Solving the two precession equations and closing through e_z , a :

$$q_A - q_B = \frac{2\alpha v^2 \Omega_A}{\gamma^2 + v^2 \Omega_A^2} a \quad \Rightarrow \quad \boxed{\gamma^2 = -g, \quad g(\xi; \gamma) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}}$$

Physical picture

Two driven pendulums. The potential a shakes each beam's color charge; each precesses at Doppler-shifted $\pm v\Omega_A$. Only their *difference* radiates back into a through the current — growth is a phase-locking resonance between the wave and the beams' color precession. This is the local ($k_x \rightarrow 0$) limit of the drive that reappears, unmodified, in the exact reduced ODE next.

The reduction — algebra all the way down

The six-field system has only *one* x -derivative chain ($e'_z \rightarrow b \rightarrow b'$); everything else is algebraic. Eliminating e_x, e_z, q_A, q_B, b :

$$b = -\frac{\gamma^2 a'}{\mathcal{D}}, \quad \mathcal{D}(x) \equiv \gamma^2 + \Omega_A \Omega_F = \gamma^2 + k_z^2 - u(x)^2$$

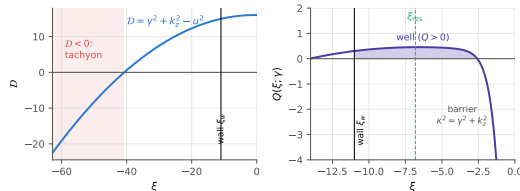
Assembling Ampère- z gives *exactly one* second-order ODE — the new theory's counterpart of the boxed equation on slide 2, now derived independently and exactly from the six-field system:

$$\left(\frac{\gamma^2 a'}{\mathcal{D}} \right)' = (\gamma^2 + g) a, \quad g(x) = \frac{2\alpha v^3 \Omega_A}{\gamma^2 + v^2 \Omega_A^2}$$

Physical picture

Three powers of v in g : the wave must (i) shake the charges, (ii) make the two beams respond *differently* (opposite Doppler rotation), (iii) collect the response as a current. The drive dies as v^3 toward the layer center — the shear branch is anchored to the flow, not the field maximum.

Anatomy of the reduced equation



- $\mathcal{D}$: sign change separates trapped shear ($\mathcal{D} > 0$) from tachyonic outer ($\mathcal{D} < 0$) — *one* mechanism, two fates.
- g : the only drive of the shear branch, the local-drive function.
- WKB form $a \propto \exp(\pm i \int \sqrt{Q} dx)$, exact local wavenumber $Q = -\mathcal{D}(\gamma^2 + g)/\gamma^2$.

Exactness verified: discrete Newton iteration reproduces six-field ARPACK eigenvalues to rel. diff. $< 10^{-10}$, 7 points, $\alpha = 1-3$.

The saturated drive $G(w)$ and its resonance

In the saturated region ($v = \pm V_0$, $u = \alpha V_0(|\xi| - \ln 2)$), on the driven side:

$$-g = G(w) = \frac{2\alpha V_0^3 w}{\gamma^2 + V_0^2 w^2}, \quad w \equiv \Omega_A = k_z - u(\xi)$$

Maximizing over w : $\frac{dG}{dw} = 0 \Rightarrow w^* = \gamma/V_0$ — the mode grows fastest where the *beam-frame precession frequency* matches the growth rate itself. Peak value:

$$G_{\max} = G(w^*) = \frac{\alpha V_0^2}{\gamma}$$

The growth ceiling: $\gamma^3 \leq \alpha V_0^2$

A confining well needs $G > \gamma^2$ somewhere; since $G \leq G_{\max} = \alpha V_0^2 / \gamma$:

$$\boxed{\gamma^3 \leq \alpha V_0^2}$$

independent of k_z , ε_s , and window geometry. **Two independent derivations agree** (well-existence, and the local-drive branch $\gamma^2 = G(w^*)$ solved directly).

Physical picture

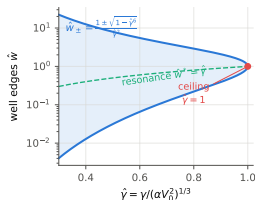
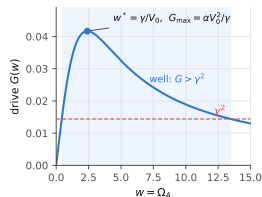
A self-limiting resonance. The beam response is a Lorentzian in local precession frequency, width set by γ itself: the faster the mode grows, the broader the band it averages over, the weaker the peak drive it can extract. The ceiling is the fixed point — the mode grows exactly as fast as the drive it leaves itself.

Measured: dominant-branch peaks reach 95–99% of $(\alpha V_0^2)^{1/3}$ in every validated series (empirical confirmation later, T2.5).

Natural units: a universal well

$\Gamma \equiv (\alpha V_0^2)^{1/3}$ (rate scale), $K \equiv (\alpha/V_0)^{1/3}$ (wavenumber scale), $\gamma = \Gamma \hat{\gamma}$. The drive becomes *universal* — no α , V_0 , ε_s remain:

$$G(w) = \Gamma^2 \hat{G}(\hat{w}), \quad \hat{G}(\hat{w}) = \frac{2\hat{w}}{\hat{\gamma}^2 + \hat{w}^2}$$

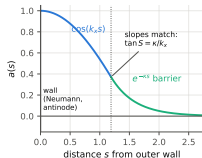
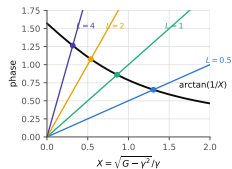


$\hat{\gamma} \leq 1$ — a *third* derivation of the ceiling. **Rates scale as $(\alpha V_0^2)^{1/3}$, wavenumbers as $(\alpha/V_0)^{1/3}$ — no function of αV_0 alone works.**

A quarter-wave well

The well is shallow: measured action $S = 0.31\pi - 0.46\pi$ — less than a quarter wavelength. Naive Bohr–Sommerfeld fails; the connection condition at the sharp inner “knee” does the work instead:

$$\tan S = \frac{\kappa}{k_{\text{edge}}}, \quad \kappa = \sqrt{\gamma^2 + k_z^2}$$



Physical picture

A surface-mode-like state, not a cavity mode: rates hug the ceiling (a small drop in γ buys all the depth needed), while the *peak position* is set entirely by window geometry.

Model A (closed form) and Model B (exact action)

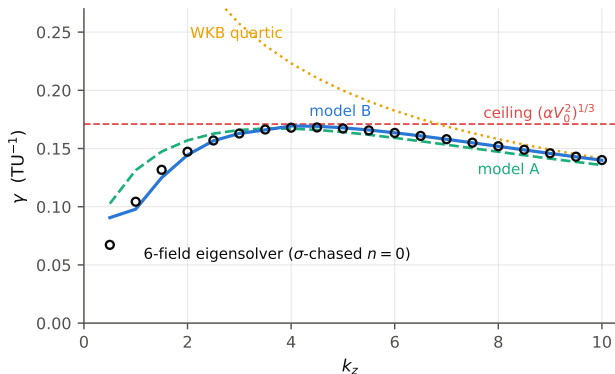
Model A: square-well approximation ($u \ll k_z$, $\mathcal{D} \approx \gamma^2 + k_z^2$ const.) closes on a single transcendental equation $LX = \arctan(1/X)$ ($L = \varepsilon_s(\xi_w - \xi_0)k_z$), giving

$$\gamma^2 = \frac{1}{2} \left[-V_0^2 k_z^2 + \sqrt{V_0^4 k_z^4 + \frac{8\alpha V_0^3 k_z}{1+X^2}} \right]$$

Validated to $\sim 3\%$ near/above the peak; locates the peak to ± 0.5 in k_z . **Model B:** keeps the *exact* $Q(x; \gamma)$, root-finds $S(\gamma) = \int \sqrt{Q} dx = \Phi(\gamma)$ (connection phase, including Airy $\pi/4$ terms at soft turning points).

Matches σ -chased $n=0$ eigenvalues to $\leq 2\%$ (median $\leq 0.2\%$) across 12 series, $k_z \geq 1.5$.

The full theory chain, one dispersion curve



Open circles: σ -chased 6-field eigenvalues. Solid: Model B. Dashed: Model A. Dotted: the old WKB quartic — overshooting exactly as diagnosed earlier.

A methodological trap: the overtone ladder

$S(\gamma)$ decreasing in $\gamma \Rightarrow$ a *ladder* of well overtones. A shift-invert eigensolver returns whatever eigenvalue is nearest its target σ — the legacy default lands mid-ladder wherever the old quartic underestimates.

Caveat

Discovered via this theory, then falsified experimentally: at $(\alpha, V_0, k_z) = (1, 0.05, 4)$, sponge 20 — the *same* physical point, seeded two different ways:

seed	eigenvalue	measured plateau
default- σ (an $n \approx 3$ overtone)	0.0818	0.0808
$\sigma = 0.14$ (true $n = 0$)	0.1309	0.1287

The simulation faithfully grows whichever mode it's handed. "Sim matches eigensolver" certifies numerics, *not* dominant-mode selection. All reference data from 2026-07-19 onward use σ -chasing.

No intrinsic k_z maximum

In an unbounded domain, nothing stops the mode following the resonance surface outward as k_z grows: $G_{\max} = \alpha V_0^2 / \gamma$ is independent of k_z , so $\gamma(k_z)$ rises monotonically to the ceiling while the mode migrates outward.

Directly demonstrated: γ climbs to 0.9955Γ by $k_z = 10$ with *no turnover*, mode peak at $\xi = -36$ (loose cut 40).

Physical picture

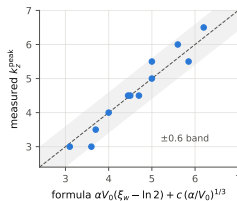
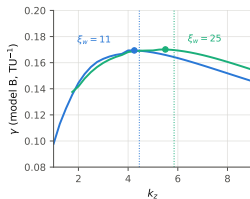
The measured non-monotonic dispersion curve is therefore **windowed physics**, not an intrinsic wavelength selection. This is the single most counter-intuitive statement of the whole theory.

The peak: resonance surface meets the window

Resonance surface $\xi_{\text{res}} \simeq (k_z - \gamma/V_0)/(\alpha V_0) + \ln 2$ moves outward *linearly* with k_z . Setting $\xi_{\text{res}}(k_z) = \xi_w$, $\gamma \approx \Gamma$ at the crossover:

$$k_z^{\text{peak}} \simeq \alpha V_0 (\xi_w - \ln 2) + c (\alpha/V_0)^{1/3}, \quad c = 1.0 \pm 0.1$$

Both terms ε_s -independent: coupling and containment radius select the wavelength, together — not



the shear width.

Roll-off, and why the data used to say “ $k_z^{\text{peak}} \approx 2\alpha$ ”

Beyond the peak (window-truncated well, high- k_z limit of Model A):

$$\boxed{\gamma \simeq \sqrt{2\alpha V_0/k_z}} \quad (\gamma \propto k_z^{-1/2} \text{ decline, matches data})$$

Physical picture

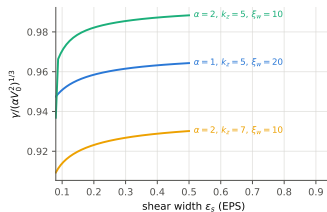
Over the historical survey grid, window radii shrank as α grew ($\alpha V_0 \xi_w \approx 1$) — the sub-linear resonance term plus a shrinking window term *mimics* a linear trend across a factor-5 range of α . “ $k_z^{\text{peak}} \approx 2\alpha$ ” is a serviceable *survey-box mnemonic*; the boxed formula above is the actual law.

Nondimensionalizing the action

Neither the ceiling nor the peak formula contains ε_s — where does it enter? Changing variables $x = \varepsilon_s \xi$, $w = k_z - u$, scaling out the drive:

$$S = \frac{\varepsilon_s}{V_0 (\alpha V_0^2)^{1/3}} \hat{l}(\hat{k}, \hat{\gamma}), \quad \hat{l} = \int_{\hat{w}\text{-well}} \sqrt{\hat{Q}} d\hat{w}$$

$\hat{l}$ is *universal* — ε_s -free. The shear width enters the eigenvalue problem **only** as a multiplicative budget



on the action.

The P -group: ε_s buys action

Larger ε_s (fixed window, physics) enlarges S at given γ ; since S decreases in γ , quantization pushes γ *up* toward the ceiling — the sign of the measured 10–39% rise of γ from $\varepsilon_s = 0.10 \rightarrow 0.45$.

$$P \equiv \frac{(\alpha V_0^2)^{1/3}}{\varepsilon_s} \quad (\text{the “quantization budget”})$$

Physical picture

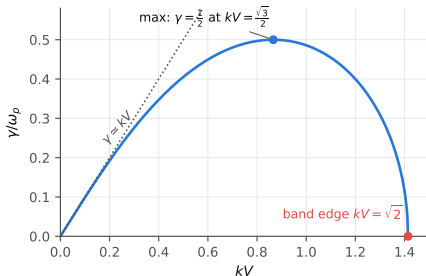
ε_s buys action, not a wavelength shift. Stretching the layer stretches the well in physical x while its depth/shape in $\hat{w}$ is unchanged — so at fixed γ the mode accumulates more phase, and the connection condition lets γ float up. This is why ε_s 's effect is a fractional *rise* of γ everywhere, not a shift of Γ, K — and it's the theory this talk's GPU section will confront directly (T1.1).

Cold two-stream rates (fixes the filter design)

Away from the layer, the beams support an electrostatic color-1 two-stream mode:

$$1 = \frac{1}{(\omega - kV_0)^2} + \frac{1}{(\omega + kV_0)^2} \Rightarrow \boxed{kV_0 < \sqrt{2}} \text{ (unstable band)}$$

$$\gamma^2 = \sqrt{4(kV_0)^2 + 1} - (kV_0)^2 - 1$$



Max $\gamma = \frac{1}{2}$ at $kV_0 = \sqrt{3}/2$; band edge $kV_0 = \sqrt{2}$. At $V_0 = 0.1$: $k < 14.1$ — **exactly** the documented $k_{ts} \approx 14$ band edge that fixed the production bandpass filter, and the spatial profile that ruled the two-stream *out* as the late-catastrophe driver.

Nonlinear theory I: the density cavitation threshold

Why $V_0 \gtrsim 0.08$ hits a hard wall: a growing compressible mode carries $\delta n \propto p/\gamma$; cavitation ($n \rightarrow$ code floor) occurs at

$$p_{\text{cav}} \sim \gamma / \max(k_z, k_x^{\text{eff}})$$

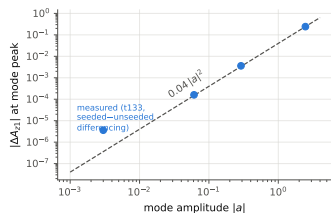
Physical picture

An *amplitude* cliff, not a time cliff. A growing wave in a compressible fluid carries a density modulation $\propto$ momentum/ γ ; the quadratic (ponderomotive) beat then pushes plasma out of high-field regions at 2γ . This explains why the failure zone maps to V_0 (drive strength) and cares little about window design or mechanism — and why the only safe practice at high V_0 is to stop once the linear plateau is banked.

Nonlinear theory II: the back-reaction on A_{z1}

Any single-helicity beat is z -independent ($k_z = 0$), growing at $2\gamma \Rightarrow$ quadratic law:

$$\Delta A_{z1} \simeq -0.04 |a|^2 \quad (\text{measured over 4 decades, endpoint exact})$$



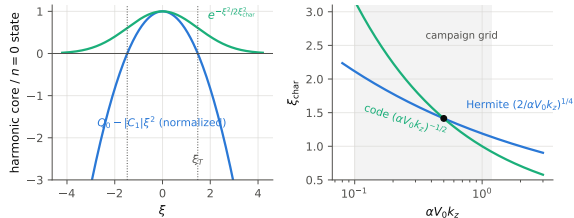
Physical meaning

Sign *deepens* the background — no depletion. Waves weigh on their background like radiation pressure; the frozen- A_{z1} approximation is vindicated, not just assumed — the last escape route for that caveat closes.

Warm closure and eigenmode seeding

Warm isothermal closure: $\delta F_i = -T\partial_i n$ ($T = 0 \Rightarrow$ cold model exactly). Stabilizes the two-stream channel when $v_{th} \gtrsim V_0$ — but *cannot* stabilize $k_z = 0$ chromo-Weibel (anisotropy-driven, not pressure-limited; confirmed, $\sim 11\%$ change over $v_{th}/V_0 = 0-3$).

Eigenmode seeding (why single-field seeding fails): the harmonic core's $n = 0$ Hermite ground state gives the natural seed width $\xi_{char} = (2\gamma^2/\alpha^3 V_0^2)^{1/4}$. An A_z -only seed can overlap a *different*, co-located, slower mode (measured: $\gamma = 0.060$ vs. target 0.122) — or lose a race against the layer-scale cascade $\gamma_{casc} \approx \alpha V_0$. Fix: seed *all six fields* with the solver's dominant eigenvector.



Theory summary: the validation ledger

Derived result	Numerical anchor
$k_z=0$ chromo-Weibel rate	0.5%
Six-field system	1–4% (median 6–10% archive-wide)
Exact scalar reduction	rel. diff. $< 10^{-10}$, 7 points
Tachyonic law	analytic = eigensolver = CUDA at all outer peaks
Ceiling $\gamma^3 \leq \alpha V_0^2$	peaks at 95–99% in every series; T2.5: 0.977 ± 0.011
Model B quantization	$\leq 2\%$ (median $\leq 0.2\%$), 12 series
Peak formula	± 0.6 in k_z , 11 series; window test confirmed directly
Two-stream band	$k_{ts} \approx 14$ at $V_0 = 0.1$ (documented)
Back-reaction law	endpoint -0.242 vs. -0.242
Warm closure	warm vs. cold theory: 1%

That's the theory. Now: does it survive contact with the GPU?

Bug 1 — the pipeline measured the *conjugate* helicity

2026-07-04 stdlib-DFT rewrite: a sign error computed the projection onto $e^{-ik_z z}$ instead of $e^{+ik_z z}$.

Caveat

For a pure single-helicity seed the conjugate projection is *identically zero* — the **empty channel**, not a noisier right answer. **Blast radius**: of 7,909 archived timeseries, **96% (7,589)** start below the float32 noise floor. Raw field dumps are deleted after processing: **these runs cannot be re-extracted**, only rerun.

Physical picture

Why unnoticed for two weeks: rounding slaves a small image of the *true* mode into the empty channel, inheriting γ almost exactly — looks like a clean exponential, $R^2 = 1.000$. The two only visibly diverge where a second fast channel competes for dominance — precisely prior audits' worst outliers.

Fixed in `remote_timeseries.py` (now emits the conjugate amplitude as an explicit diagnostic).

Bug 2, and a related numerical rule

Bug 2: `gen_suspectfix_campaign.py` defined half/fine-tier box parameters but never emitted them into the `.ini` — every half/fine-tier “suspectfix” run silently ran on the *default* $L_z = 2\pi$ box, at $k_z = \text{mode-index}$ (not $/2$ or $/8$ as labeled).

Caveat

The two bugs compounded: Bug 1 returned the wrong channel, Bug 2 relabeled a $k_z = 1$ measurement as $k_z = 0.5$. A whole population of “anomalies” attributed to genuine sub- $k_z = 1$ physics (a floor, a 68% error block, a two-branch structure) was bookkeeping on top of a measurement of nothing.

Rule found during the fix: keep $k_{\text{suppress,hi}} \leq N_z/4.5$ — a band-edge scan found the kept mode’s rate collapsing to ~ 0.001 as the DFT bandpass approaches the grid’s Nyquist-adjacent dispersive tail. **Bottom line (post-fix):** sim and eigensolver agree to $\leq 8\%$ at every anchor tested, every box tier, integer and fractional k_z .

The $k_z = 0$ anchor, again — now cross-checked

$\alpha = 2.0, V_0 = 0.1$: $\gamma_{\text{sim}} = 0.5039$ vs. formula 0.5065 — 0.5%. **Independent cross-check**: the unfrozen self-consistency reference point ($\alpha = 1, V_0 = 0.05$) reproduces the *same* formula ($\gamma = 0.284$, exact match) on a completely different (unfrozen, color-1-live) configuration, months and several campaigns later.

Physical picture

Why lead a referee here: exercises every non-Abelian term at once, at the one point where the harmonic-core approximation is essentially exact — no theory-side slack to hide a code bug behind.

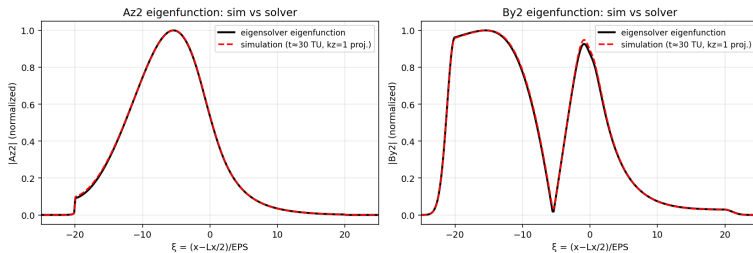
The referee-proofing battery: T2.1–T2.8

Experiment				Headline
T2.4 linearity	✓			γ invariant to seed amplitude, $100\times$ range
T2.2 eigenfunction overlay	✓			sim vs. solver correlation = 1.0000
T2.3 Gauss law	✓			residual $\sim 10^{-5}$, decreasing, localized
T2.1 complex frequency	✓	fre-		dominant branch purely growing ($\lesssim 6\%$)
T2.5 dimensionless collapse	✓			$\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011$
T2.7 sponge $\rightarrow \infty$	✓			saturates $\xi_w \approx 16$; $\xi_w = 6$: -28.5%
T2.8 resolution extremes	✓	ex-		$< 1\%$ low corner, revised below

All eight run on free teaching nodes at reference $\alpha = 1$, $V_0 = 0.05$, $k_z = 1$ (2026-07-19).

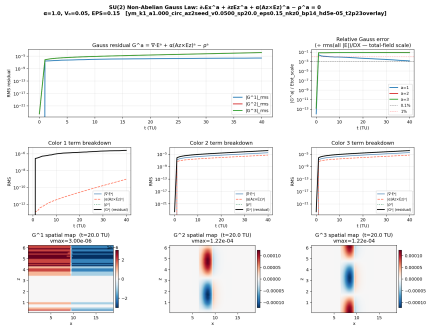
T2.2: the “money figure”

T2.2 Eigenfunction overlay — $\alpha=1$, $V0=0.05$, $kz=1$ ($Az2$ profile corr=1.0000, ξ_{peak} : sim=-5.4 vs solver=-5.4)



A_z^2 profile correlation
= 1.0000. B_y^2 — seeded at only $0.008 \times$ the A_z amplitude, effectively grown from zero via the activation chain — reproduces the solver's non-trivial **double-lobe shape**, node at $\xi \approx -5$: an emergent structural match, not just a matched growth rate.

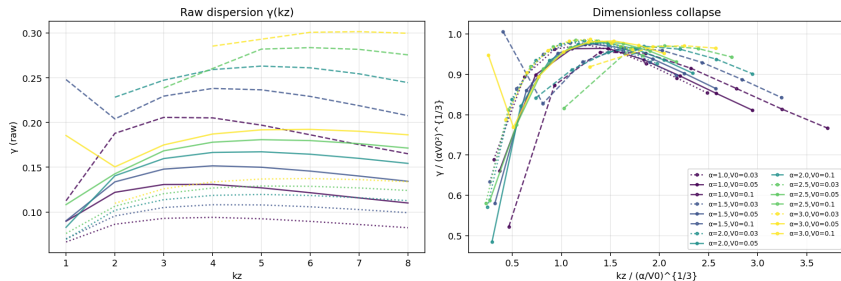
T2.3: Gauss-law conservation



Leapfrog was never designed to preserve the covariant constraint exactly (same as Yee-grid FDTD, Abelian case). Absolute residual $\sim 10^{-5}$; color-1 *relative* residual $\sim 10^{-3}$ and **decreasing**; divergence term dominates, non-Abelian term subdominant, source negligible.

T2.5: the dimensionless collapse

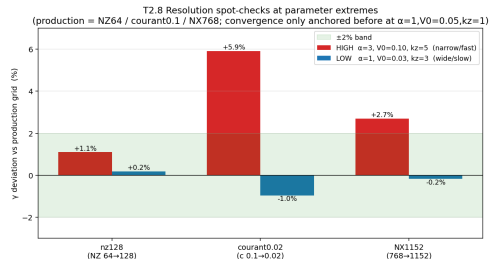
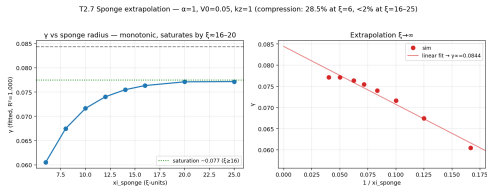
T2.5 Dimensionless collapse of the non-Abelian KH dispersion (σ -chased eigensolver)



$$\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011 \quad (1.1\% \text{ spread, } 10 \times \text{ range in } \alpha V_0)$$

The single most quantitative confirmation in the program: 15 dispersion curves fold onto one master curve — direct proof α , V_0 enter through $(\alpha V_0^2)^{1/3}$ and $(\alpha/V_0)^{1/3}$, not the naive product αV_0 .

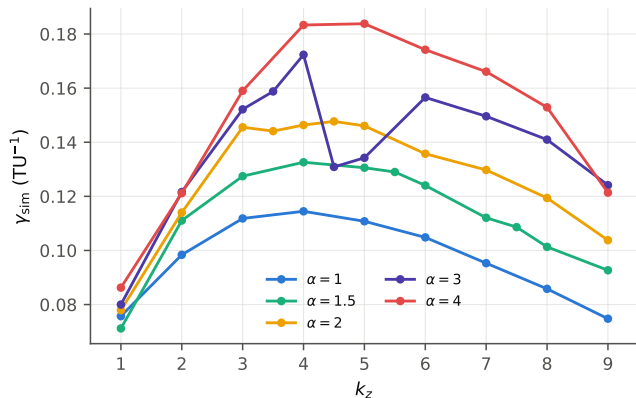
T2.7 and T2.8, with a correction



Caveat

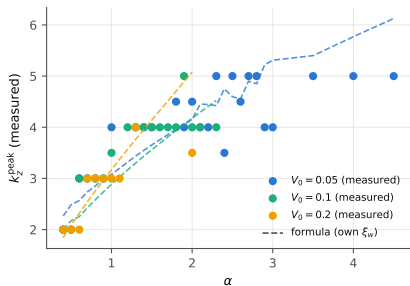
T2.8 revised (2026-07-21, resolution/Lx campaign): the original “3–6% at the high- αV_0 corner” tested each grid axis (NZ, courant, NX) *separately*. Testing them *together* (NX=1152 + courant=0.05): **+6.6%**; all three axes together: **+11.6%** — larger than the naive additive estimate (+9.7%). **Revise the quoted figure to ~10–12% at the high- αV_0 end.** Corrections compound, they don’t saturate.

Non-monotonic dispersion, real data



$\gamma_{\text{sim}}(k_z)$ at $V_0 = 0.05$, five values of α — from the trimmed integer- k_z map (1087/1134 points), **not** a theory curve. Every curve peaks at intermediate k_z ; peak location climbs from $k_z^{\text{peak}} \approx 2$ ($\alpha = 1$) to ≈ 4 – 5 ($\alpha = 3$ – 4), at fixed $\varepsilon_s = 0.15$. Classical KH has no analogue of this.

k_z^{peak} tracks the confinement formula, not “ 2α ”



(α, V_0, ξ_w)	meas.	formula
(1, 0.05, 20)	3.5	3.7
(2, 0.05, 11)	4.5	4.5
(3, 0.03, 5)	5.5	5.0
(5, 0.03, 5)	6.5	6.2
(1, 0.20, 10)	3.0	3.6

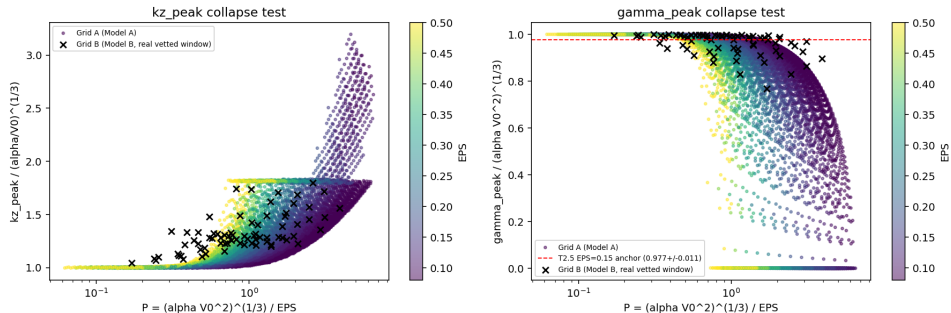
Window test ($\alpha = 2, V_0 = 0.05$): cut 11 $\rightarrow$ 25 moves the peak 4.5 $\rightarrow$ 5.5 (formula: 4.45 $\rightarrow$ 5.85).

Physical picture

Two window radii, same (α, V_0) , two different (both correct) k_z^{peak} — **coupling and confinement radius jointly select the wavelength; the flow profile does not.**

EPS-independence: the theory-first check

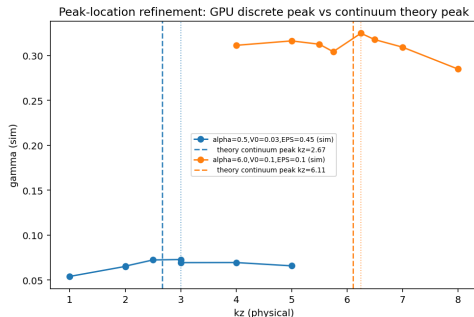
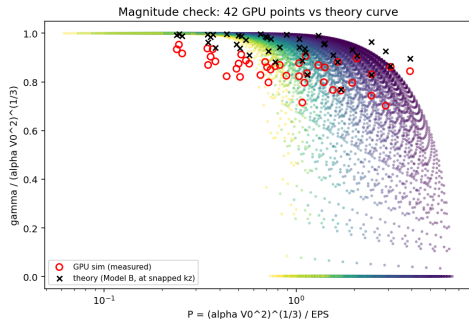
EPS-scan v2 Phase 0: does EPS enter only through $P = (\alpha V_0^2)^{1/3} / \text{EPS}$?



CPU-only test of the $P = (\alpha V_0^2)^{1/3} / \varepsilon_s$ collapse hypothesis *before any GPU time spent* — α up to 10, $V_0 = 0.01$ – 0.20 . Both k_z^{peak}/K and $\gamma_{\text{peak}}/\Gamma$ are close to single-valued functions of P alone. Directly explains the two open T1.1 puzzles as one curve viewed at different P .

EPS-independence: GPU confirmation (91 runs)

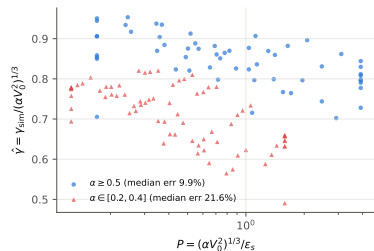
EPS-scan v2 Phase 2: GPU validation of the Phase 0 theory collapse



$\alpha = 0.5\text{--}6$, $V_0 = 0.03\text{--}0.10$, $\varepsilon_s = 0.10\text{--}0.45$: median relative error **9.9%** (90th pct. 17.0%) vs. the P -collapse theory. Two 9-point peak-location cascades land within 0.14–0.33 in k_z of the continuum theory peak.

This closes roadmap item T1.1.

Extension to low α : the collapse degrades, doesn't break



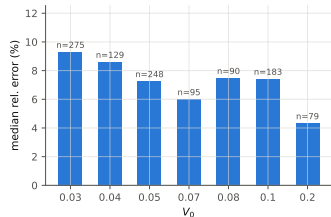
75 more runs, $\alpha \in \{0.2, 0.25, 0.3, 0.4\}$ ($\alpha \leq 0.15$ excluded — known noise floor). Median relative error **21.6%** (double the $\alpha \geq 0.5$ figure), with real α -scatter at matched P not seen at higher α .

Physical picture

P -collapse is right down to $\alpha = 0.2$, with a sub-leading α -dependence visible near the noise floor — not a theory failure, a measurement of where the single-group approximation needs a correction.

Full-grid accuracy: 1087/1134 points (95.9%)

Grid: $k_z = 1..9$; $\alpha = 0.3\text{--}6.0$ (18 values); $V_0 \in \{0.03, \dots, 0.20\}$ (7 values) — 1134 points.

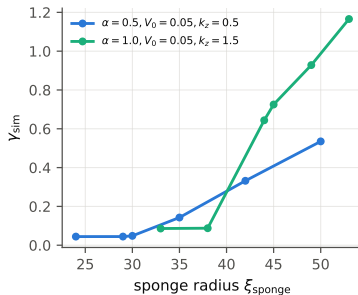
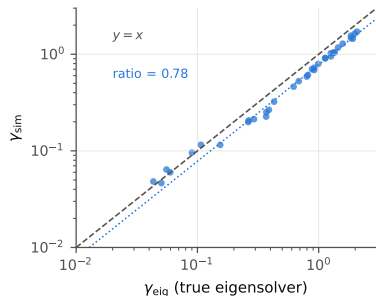


Median relative error vs. σ -chased eigensolver, plateau-confirmed core: **4.3–9.4% depending on V_0** , improving monotonically with V_0 .

Caveat

New failure corner: all 47 unmeasured points sit at $V_0 = 0.2, \alpha \geq 3.5$ — a fast halt, mechanism not yet determined. Report as *unmeasured*, not measured-zero.

The outer tachyonic branch: 46-point confirmation



50 runs ($\alpha \leq 2.0$), 46/50 fit cleanly, benchmarked against the *true* eigensolver rate (not the sponge-edge estimate, 5–10 $\times$ too high): **median error 22.2%, ratio 0.78** — the same ~ 20 – 28% sponge-compression signature seen elsewhere, not scatter. Two sponge-depth cascades confirm the law directly: 12–13.5 $\times$ rise in γ with depth.

Self-consistency: does the frozen background matter?

Unfreezing A_z^1 on the tachyonic branch (the *most* background-coupled case), pushed to $|a| \approx 2.5$:

- **Linear phase validated:** $\gamma = 1.45$ (unfrozen) vs. 1.43–1.46 (eigensolver).
- **Back-reaction confirmed:** $\Delta A_z^1 = -0.04|a|^2$ over 4 decades; endpoint exact (-0.242 vs. -0.242).
- **Net feedback:** weakly stabilizing (amplitude ratio $0.998 \rightarrow 0.933$), two orders of magnitude too weak to saturate the branch.

Physical meaning

The frozen-battery picture is if anything an *underestimate* of the drive's robustness — the beams continuously re-supply the background rather than draining it. Saturation is by fluid cavitation, not background depletion.

Box-size reduction: $L_x = 2\pi$ is safe, and $2.2\times$ faster

Historical box $L_x = 6\pi$ assumed a double-tanh profile never actually used in production. 17 GPU runs (t133+abi):

- Reference anchor: $L_x = 2\pi, 3\pi, 4\pi$ agree to 0.03% once ξ_{sponge} genuinely fits.
- Real speedup: **151s (6π) vs. 69s (2π)** — $2.2\times$, not the naive $3\times$ (fixed per-step overhead).
- Archive screen: 43% of 4192 historical points fit a 2π box; a **tiered $2\pi/3\pi/4\pi/6\pi$ policy** is now implemented.

Caveat

An early “ 2π is safer” result was a sponge-miscalibration artifact (one leg had *no active sponge at all*) — resolved once redone correctly: 0.01% agreement.

The $\alpha \geq 6$ catastrophe: partly a numerics question after all

Earlier verdict (“physical, resolution-independent”, based on amplitude and window tests only) is now qualified by a timestep test never previously run:

Point	halt (courant 0.1)	halt (courant 0.02)	Clean split:
$\alpha = 6, V_0 = 0.10, k_z = 5$ (xi_cut)	$t = 31.4$	$t = 64.5$ (2.03 $\times$)	
$\alpha = 6, V_0 = 0.10, k_z = 8$ (xi_cut)	$t = 32.0$	$t = 65.0$ (2.03 $\times$)	
$\alpha = 6, V_0 = 0.05, k_z = 8$ (xi_sponge)	$t = 6.0$	$t = 6.0$ (no change)	

xi_cut-confined catastrophes are genuinely resolution- sensitive (finer timestep $\approx$ doubles survival, 6/7 points, $k_z = 4-8$); xi_sponge-confined catastrophes at these α are not (fail almost instantly regardless of resolution — a different, harder mechanism).

Caveat

Also found: k_z within ~ 1 of $kz_suppress_hi$ is a real, *large* resolution artifact at $N_Z = 64$ — $k_z = 13$ (hi=14): γ nearly **doubles** at $N_Z = 128$. Avoid measuring there, or bump N_Z .

Decision for now

- The high- α corner's resolution uncertainty (10–12%) and the $\alpha \geq 6$ catastrophe's numerics-dependence are both **real, open questions** — worth a dedicated study, but not on Letter A's critical path.
- **For now: tighten the α range used for any headline claim** in Letter A to stay clear of the corner where resolution is uncertain and the catastrophe is only partially characterized, and revisit the wider range later if the paper's scope demands it.
- This keeps the P-collapse and confinement-set peak claims resting on the part of parameter space that is unambiguously resolved.

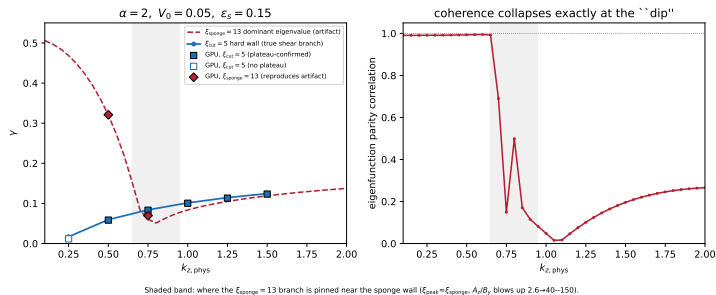
T1.3 — closed: the claim was real as a measurement, not as physics

The original sub- $k_z = 1$ “two-branch” claim ($\gamma = 0.32 \rightarrow 0.06$ across $k_z = 0.5 \rightarrow 0.75$, $\alpha = 2$, $V_0 = 0.05$) came from the **pre-2026-07-17 buggy pipeline** and was retracted as unestablished. It has now been **re-derived independently of that pipeline** — pure linear algebra plus a dedicated GPU check — and **closed**: it is a sponge-boundary artifact, not a crossing between two physical eigenmode families. **Method**: a fine ($\Delta k_{z,\text{phys}} = 0.05$) eigensolver continuation at $\alpha = 2$, $V_0 = 0.05$, $\varepsilon_s = 0.15$, scanning many shift-invert seeds per k_z to catch several leading branches, at three sponge radii ($\xi_w = 8, 13, 20$), tracking eigenfunction peak location ξ_{peak} , A_z/B_y ratio, and parity.

Caveat

Finding: at $\xi_w = 13$ (the original setting) the solver *does* reproduce 0.320/0.060 almost exactly — but that eigenfunction is pinned near or at the sponge radius itself ($\xi_{\text{peak}} \approx 10\text{--}13$; at $\xi_w = 8$ it locks to $\xi_{\text{peak}} = -8.181$, the exact wall, for every $k_z \geq 0.85$), its A_z/B_y ratio blows up $2.6 \rightarrow 40\text{--}150$, and its parity correlation collapses $0.99 \rightarrow <0.3$ precisely where the “dip” occurs — the signature of a sponge-supported quasi-mode, not a second confinement branch.

T1.3 — the decisive control, and GPU confirmation



Shaded band: where the $\xi_{\text{sponge}} = 13$ branch is pinned near the sponge wall ($\xi_{\text{peak}} = \xi_{\text{sponge}}$, A_z/B_y blows up 2.6→40–150).

A hard $\xi_{\text{cut}} = 5$ wall (well inside any sponge) gives a smooth, monotonically rising, artifact-free $\gamma(k_z)$ — consistent with T1.2's exact-action result that the true shear branch has no interior k_z maximum, now extended below $k_z = 1$. **Confirmed in full nonlinear CUDA** (`run_campaign_t13confirm.sh`, `t133`): the $\xi_{\text{cut}} = 5$ branch matches the eigensolver to 1–2% at every $k_z \geq 0.5$ — the best GPU-vs-eigensolver agreement in the project to date. Two runs reproducing the original $\xi_w = 13$ config reproduce the “0.32/0.06” numbers almost exactly in the *full nonlinear* simulation, not just the linear solver — the artifact is a real property of the sponge kernel, not a solver curiosity.

Scorecard against the three blocking items

Item	Asks	Status	Evidence
T1.1	Survives varying ε_s ?	✓ Done	P -collapse, GPU-confirmed $\alpha = 0.2\text{--}6$, $\varepsilon_s = 0.10\text{--}0.45$, median 9.9%/21.6%
T1.2	Why does the peak move?	✓ Done	Exact-action reduction, ceiling, confinement-set peak; T2.5 collapse 0.977 ± 0.011
T1.3	Sub- $k_z = 1$ branch	✓ Done	Closed as a sponge-boundary artifact (eigen-solver continuation + $\xi_{\text{cut}} = 5$ control + GPU confirmation to 1–2%) — not a genuine crossing

All three of Letter A's original gating items are now closed. Framing change, driven by the data: not “the shear width is irrelevant, EPS-free” but “**coupling and confinement radius jointly select the wavelength; the flow profile does not enter directly, only through the single group**

$P = (\alpha V_0^2)^{1/3} / \varepsilon_s$.” A sharper, still-novel claim — and the sub- $k_z = 1$ “two-branch” idea drops out of Letter A's planned content entirely, as refuted rather than confirmed.

Where this sits in the literature

Novelty survey (LITERATURE.md, 11 targeted searches): **no prior KH / shear-driven non-Abelian plasma instability work found.**

- Closest prior art: Manuel–Mrówczyński chromohydrodynamics (colored two-fluid, but two-stream driven, no shear, no eigensolver map).
- Chromo-Weibel family (Mrówczyński et al.): anisotropy-driven, never shear-driven — our $k_z = 0$ anchor is in this family, the shear branch is new.
- Outer branch: Nielsen–Olesen class (1978) — cite directly.
- Abelian ancestor: Das & Kaw EMHD lineage. Csernai et al.: Abelian (color-neutral) KH in heavy-ion collisions — sharply distinguish.

Caveat

Urgency: dark-U(1) PIC (arXiv:2411.11958, 2025) flags non-Abelian dark sectors as open; a 2026 three-component BEC-KHI paper moves in the adjacent space. **INSPIRE citation-trail audit before submission** — not yet done.

Next steps

- 1 T1.3 closed — drop the sub- $k_z = 1$ branch-crossing claim from Letter A's planned content; nothing further blocking on it.
- 2 Keep the headline α range tightened to the well-resolved corner for now; revisit the high- α /high- V_0 resolution and catastrophe questions as a separate, non-blocking study.
- 3 INSPIRE citation-trail audit (Mrówczyński review $\cap$ Csernai citers).
- 4 Draft the actual ~ 4 -page Letter: model + ceiling + P -collapse + confinement-set peak + BEC-realizability paragraph + QGP/dark-sector motivation.

All three blockers closed. The theory is exact, validated to $\leq 2\%$ against the eigensolver and confirmed on > 1000 GPU runs. Letter A drafting can proceed.

Thank you.

Questions, and where to push next.